COSI Data Challenge 2022 README

October 2022

General introduction

Welcome to the first COSI Data Challenge! This is the first of many COSI Data Challenges to be released on a yearly basis in preparation for the launch of the COSI Small Explorer mission (Tomsick et al. 2019) in 2027. The main goals of the COSI Data Challenges are to facilitate the development of the COSI data pipeline and analysis tools, and to provide resources to the astrophysics community to become familiar with COSI data. This first COSI Data Challenge was funded through NASA's Astrophysics Research and Analysis (APRA) for the release of high-level analysis tools for the COSI Balloon instrument (Kierans et al. 2017), and thus the COSI Balloon model and flight data will be the focus this year. Future COSI Data Challenges will be released for the SMEX mission with increasingly more sophisticated tools and a larger range of astrophysical models and simulated sources each year. By the time we're ready to fly COSI, we will have simulated all of the main science objectives, developed the tools required to analyze each case, and have educated a broader community to perform the analyses.

Download and Install

Already have COSItools installed:

If you already have the COSItools installed on your computer, type `cosi` to navigate to the COSItools directory and activate the cosi python environment. Clone the cosi-data-challenge-1 repository.

The file requirements.txt lists all of the dependencies for running the data challenge, and they can all be installed using pip: `pip install -r requirements.txt`.

You will also need to add the cosipy-classic directory to your python path, e.g. `export PYTHONPATH=$PYTHONPATH:/path/to/COSItools/cosi-data-challenge-1/cosipy-classic`

Some of the data products are large and we are using the Git Large File Server. You will need to install git-lfs:
https://docs.github.com/en/repositories/working-with-files/managing-large-files/installing-git-large-file-storage. After Git LFS has been successfully installed, navigate to your local cosi-data-challenge-1 directory and `git lfs pull` to download all of the data products. If this step is not performed, the file names will appear locally, but they will

only be placeholders and the size will be only a few hundred kB. The total size of the data products should be 4.6 GB.

You should now be able to start a python session, or open one of the provided notebooks, and import the COSIpy_dc1 module, i.e. `from COSIpy_dc1.py import *`, without issue. If you have any issues with the initization, please contact Chris Karwin - see **Issues and Feedback** below.

New to COSItools:

Head to the feature/initialsetup branch of the cosi-setup Git repository and follow the readme guide: https://github.com/cositools/cosi-setup/tree/feature/initialsetup.

Note that when making the installation you must include the option "--extras=cosi-data-challenge-1". This installation will include MEGAlib, ROOT, Geant4, Git LFS, and all packages in the requirements.txt file. After successful installation (this will take time), you will want to activate the cosi python environment by typing `cosi`.

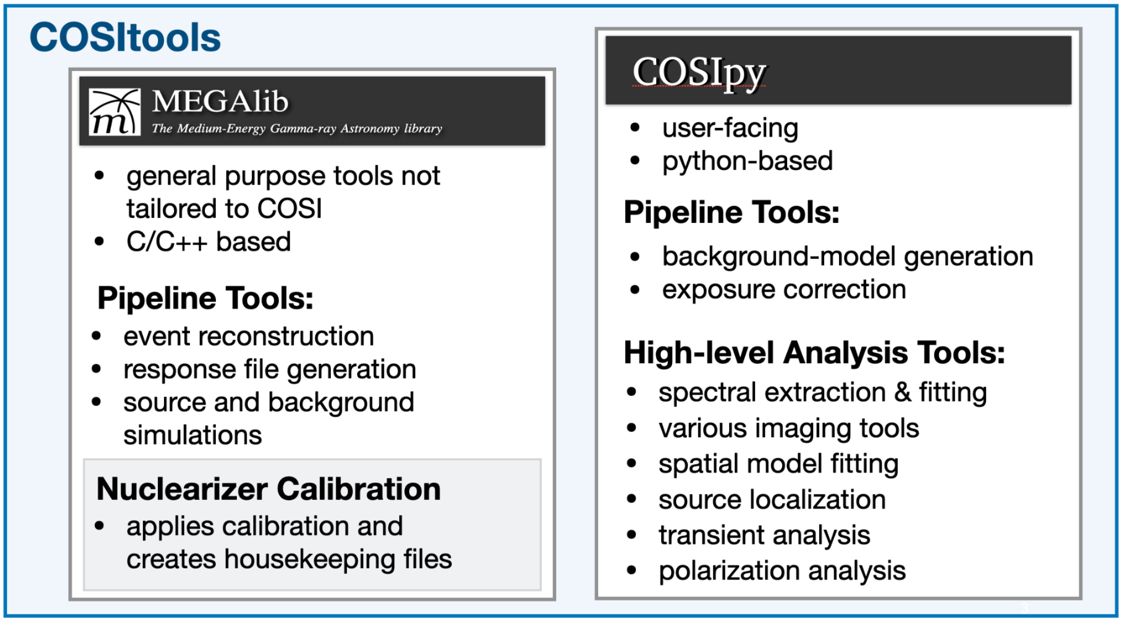
You will also need to add the cosipy-classic directory to your python path, e.g. `export PYTHONPATH=$PYTHONPATH:/path/to/COSItools/cosi-data-challenge-1/cosipy-classic`

Get Help

Since this is the first release for COSIpy there will likely be issues, and we definitely hope to get feedback from the community on what worked well, what didn't, and how can we improve for next time. Please submit a New Issue in git if you have issues with the code. If you have general feedback, or need further assistance, please reach out to the COSI Data Challenge team lead: Chris Karwin.

Getting Started

The COSI pipeline tools, COSItools, are divided into two programs (see figure below): MEGAlib and COSIpy. MEGAlib (https://megalibtoolkit.com and https://github.com/zoglauer/megalib) is the Medium Energy Gamma-ray Astronomy Library, which is a general purpose tool that is state-of-the-art for MeV telescopes. MEGAlib performs the data calibration, event identification, reconstruction, as well as detailed source simulations. There are some high-level analysis tools in MEGAlib, but most of the COSI science analysis will be performed in COSIpy. COSIpy is the user-facing, python-based, high-level analysis tool for COSI data.



This Data Challenge will serve to introduce the community to COSIpy and general Compton telescope analysis. We have prepared Jupyter Notebooks to walk the user through the analyses which are provided under spectral-fit and imaging; however, we suggest reading through the below description before attempting the notebooks.

COSIpy was first developed by Thomas Siegert in 2019 to perform 511 keV image analysis from the 2016 COSI balloon flight (Siegert et al. 2020). Since then, it has been used for point source imaging and spectral extraction (e.g. Zoglauer et al. 2021), aluminum-26 spectral fitting (Beechert et al. 2022) and Al-26 imaging, all using data from the COSI Balloon 2016 flight. These analyses and the current Data Challenge use what we refer to as “COSIpy-classic.” The team is currently working on improved response handling and streamline tools built from the bottom up, and the new and improved COSIpy will be the focus of next years’ Data Challenge! With that in mind, there are still known issues and limitations with COSIpy-classic that we will call out throughout this work.

The Simulated Data

For the first Data Challenge, we wanted to give the users a basic look at COSI data analysis, so we chose 3 straightforward examples based on the COSI science goals:

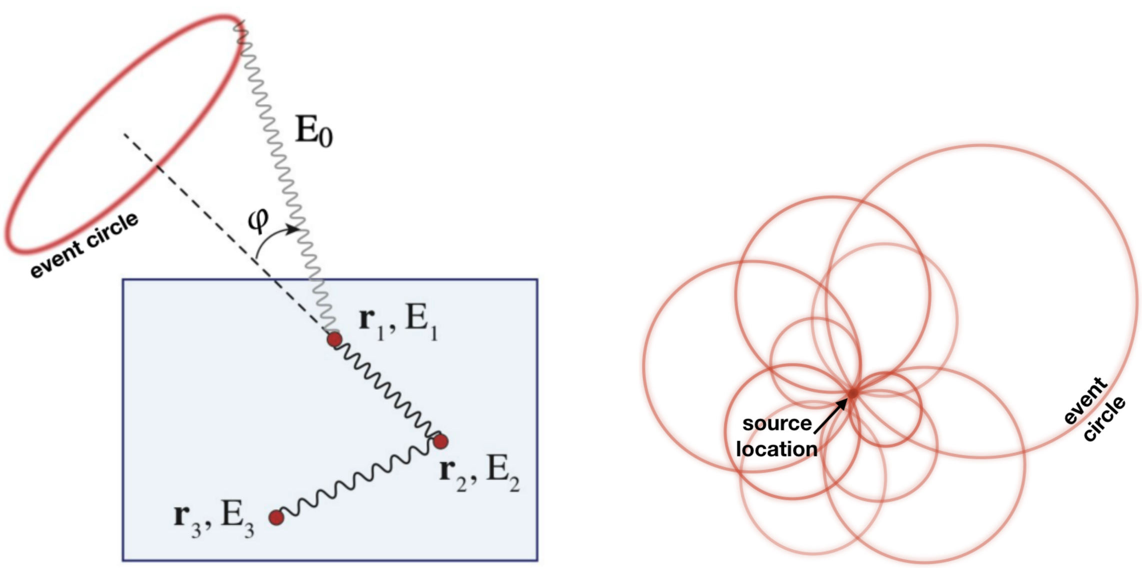
1. Extracting the spectra from the Crab, Cen A, Cyg X-1, and Vela.
2. Imaging bright point sources, such as the Crab and Cyg-X1.
3. Imaging diffuse emission from 511 keV and the Al-26 1.8 MeV gamma-ray line.

For each of these examples, we have provided a detailed description of the simulated sources and data products in the data_products directory. Each of the sources was simulated at 10x the astrophysical flux since the balloon flight had limited observation time, and because there were multiple detector failures during the balloon flight which reduced the effective area significantly.

General Compton Telescope Analysis Procedure

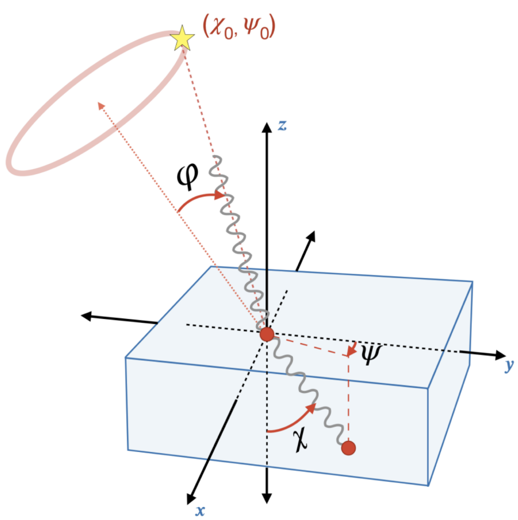
COSI is a Compton telescope, and analysis of MeV data is challenging due to the high-backgrounds and complicated instrument response. Thus, we will provide a basic description of Compton telescope data analysis here as an introduction for new users before diving into the Data Challenge analysis. For those who are new to Compton telescopes, we encourage you to read the review: Kierans, Takahashi & Kanbach 2022.

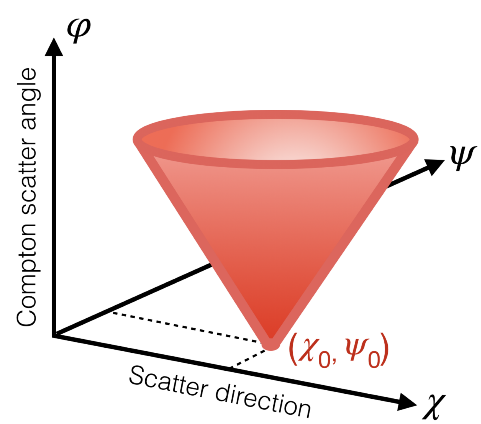
Compton telescopes provide a technique for single-photon detection, and each photon is measured as a number of energy deposits in the detector volume, shown in blue in the figure below. These interactions must first be sequenced, and then reconstructed to determine the original direction of the photon, which is constrained to a circle on the sky (a so-called Compton event circle) defined with opening angle equal to the Compton scattering angle (ϕ) of the first interaction. The classic schematic for Compton telescopes is shown in the figure below, where this example shows a photon that results in 2 Compton scattering interactions, and finally a photoelectric absorption in interaction 3, fully containing the energy of the photon in the active detector volume. The event sequencing and reconstruction all occurs in the MEGAlib tool, and we will be starting our analysis with events that are already defined by their total energy deposit in the detector (E_0), and the Compton scattering angle.



The above figure is traditionally shown for Compton telescopes, where an image of the source distribution can be realized by finding the overlap of event circles from multiple source photons. A point-source image can be recovered by performing deconvolution techniques. This List-mode imaging approach (Zoglauer 2005) is implemented in MEGAlib, and can be used for strong point sources, for example in laboratory measurements. However, this assumes a simplified detector response and cannot be used for the most sensitive analyses.

When performing analyses of astrophysical sources with high backgrounds, a more sophisticated description of the data space is required. This data space, along with the fundamentals of modern Compton telescope analysis techniques, was pioneered for the COMPTEL mission (Schönfelder et al. 1993 & Diehl et al. 1992) and is sometimes referred to as the COMPTEL Data Space, or simply the Compton Data Space (CDS). In the CDS, each event is defined by a minimum of 5 parameters. Three parameters describe the geometry of each event: the Compton scatter angle (ϕ) of the first interaction, and the polar and azimuthal angles of the scattered photon direction (χ,ψ). These angles as defined in instrument coordinates are shown in the figure below. The other two parameters in the CDS are the total photon energy (E_0) and the event time (t), but these are not always explicitly written.



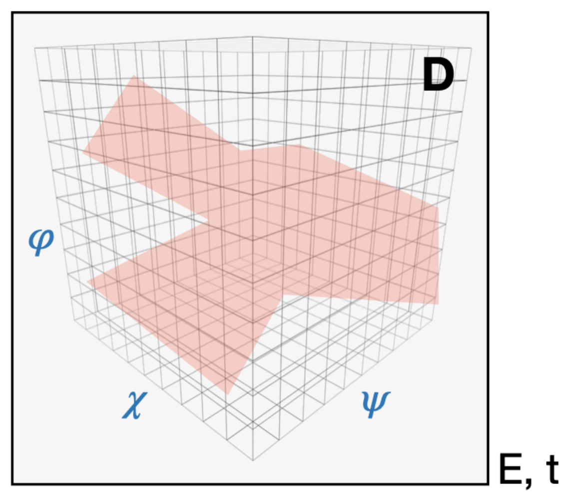


The three scattering angles (ϕ,χ,ψ) make up 3 orthogonal axes of the CDS. As photons from a point source at location (χ0,ψ0) scatter in the detector, the CDS will be populated in the shape of a cone with apex at the source location, as shown in the figure on the right. This is the point spread function of a Compton telescope. The opening angle of the CDS cone is 90º since the Compton scatter angle is equal (within measurement error) to the deviation of the scattered photon direction. An extended source will appear as a broadened cone. The more familiar Angular Resolution Measure (ARM) for Compton telescopes is a 1-dimensional projection of the width of the CDS cone walls, representing the angular resolution.

All analyses with COSIpy start with reconstructed events defined in a photon list (MEGAlib's .tra files). After reading in the data from the .tra file, the first step of any analysis is to bin the data into the CDS. For this Data Challenge, we are using 6º bin sizes for each of the scattering angles. This is because the angular resolution of the COSI Balloon instrument is ~6º at best, and smaller bins would be more computationally demanding. We are also using 10 energy bins for the continuum analyses, and 2 hour time bins for the spectral analysis and 30 min time bins for the imaging. For the narrow-line sources, such as 511 keV or Al-26, we use only 1 energy bin centered on the gamma-ray line of interest.

As a visual representation of the data (D) in the CDS, see the figure below. The three axes are the 3 scatter angles, and each bin contains the number of events, or counts, with that (ϕ,χ,ψ), shown with the red color fill (this is just a random distribution and not representative of what real data looks like in the CDS). The CDS is filled for each energy and time bin, represented by the subscript E,t in the figure. In COSIpy-classic, χ and ψ are binned into in 1145 FISBEL bins. FISBEL stands for Fixed Integral Square Bins in Equi-Longitude, and is MEGAlib's spherical axis binnings that have approximately equal solid angle for each pixel.

Compton Data Space



6º bins (30x φ bins, 1145x $\chi\psi$ FISBEL bins), 10x energy bins, 1hr time bins

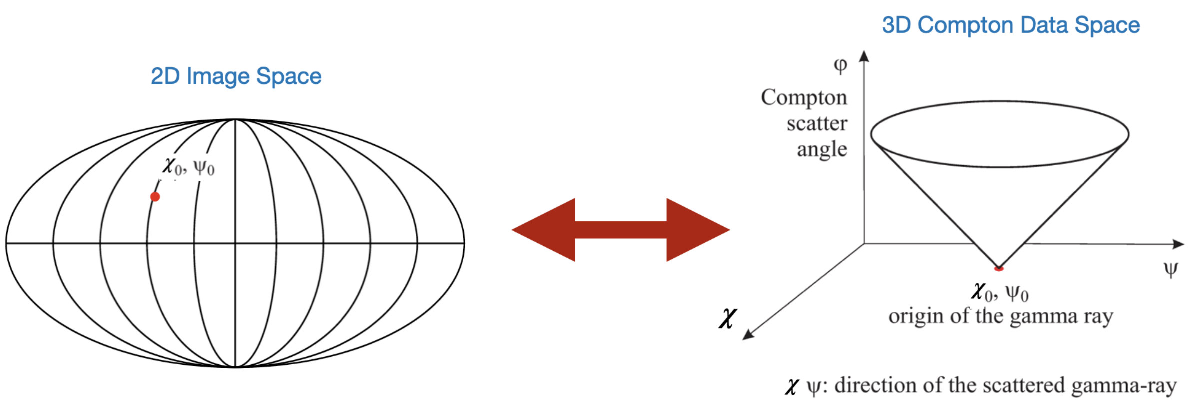
Now that we have a better understanding of the Compton telescope data space, and have our data binned in the CDS, we're ready to understand how to perform spectral analysis and imaging with COSI. There are 3 key pieces needed:

1. Response Matrix
2. Sky Model
3. Background Model

We will describe these components here and explain how they are used for general fitting procedures with COSI data. When running through the analysis notebooks pay attention to when these are initialized.

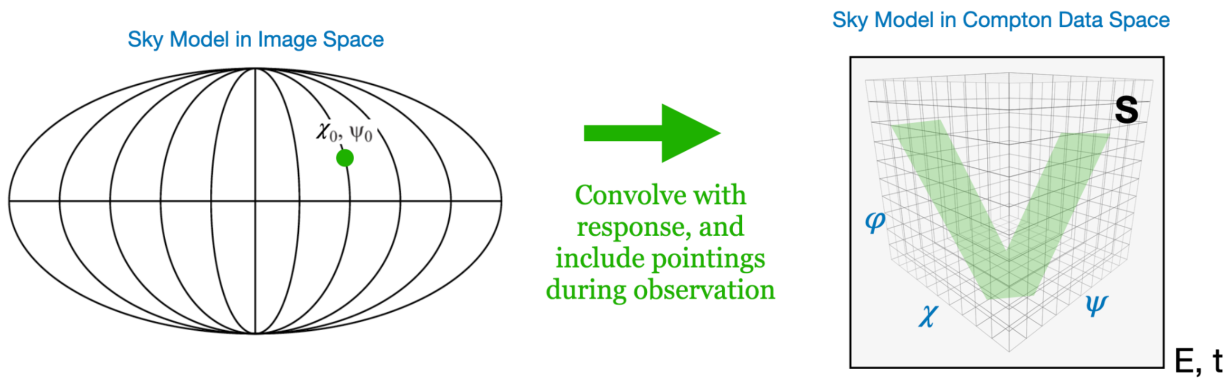
Response Matrix

The response matrix (R) for Compton telescopes represents the probability that a photon with energy E initiating from Galactic coordinates (l,b) interacts in the detector resulting in an event with measured energy E' and scattering angles (ϕ,χ,ψ). The probability distribution is normalized to match the total effective area of the instrument, which allows us to go from counts to physical parameters. The matrix that encodes this information is 2 dimensions larger (l,b) than the CDS and describes the transformation between image space and the CDS, taking into account the accurate response of the instrument. We build the response matrix through large MEGAlib simulations of an isotropic source.



Sky Model

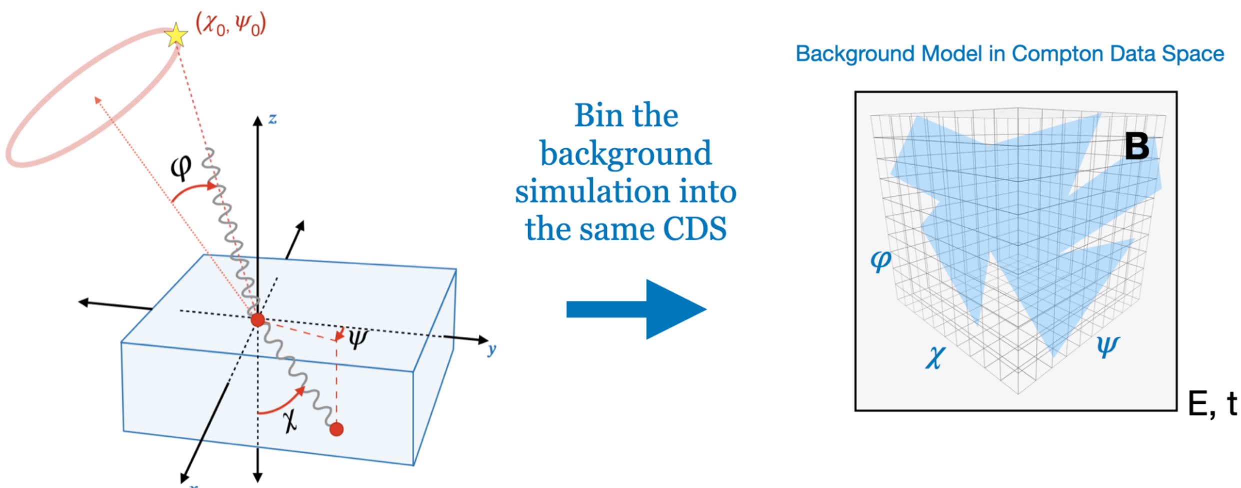
For forward-folding analyses methods, we assume a source sky distribution, referred to as the sky (or signal) model (S). The sky model in image space can be simply a point source or it can be a complicated diffuse model. The sky model also contains the spectral and polarization assumptions. This model is convolved with the instrument response matrix, and includes knowledge of the instrument aspect during observations, to determine the representation of the sky model in the CDS. For example, here we are showing this for a simple point source, and we regain the expected cone-shape in the CDS.



Background Model

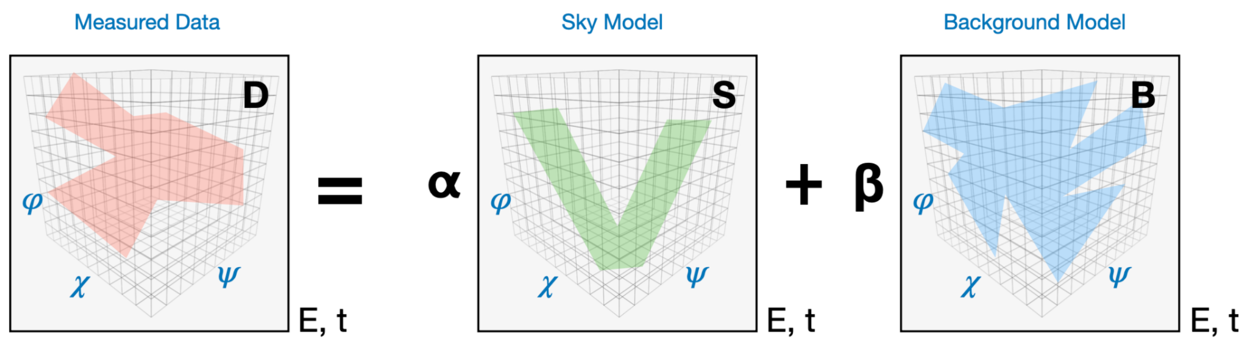
We require an accurate estimate of the backgrounds during observations. This background model (B) can be achieved in a number of ways, for example by using the measured flight data from source-starved regions. Alternatively, one can perform full bottom-up simulations of the gamma-ray background at balloon-flight altitudes, including activation. For this first Data Challenge, we are using this approach, and the simulation is further described in data_products; however, for future Data Challenges, we will be employing multiple background-model approaches.

With the background model generated from simulations, then we first have to bin it in the same CDS that we have used for the data and the source model. We have already performed this step for you, and have provided a .npz file, which is a zipped numpy array of the background simulation in the CDS.

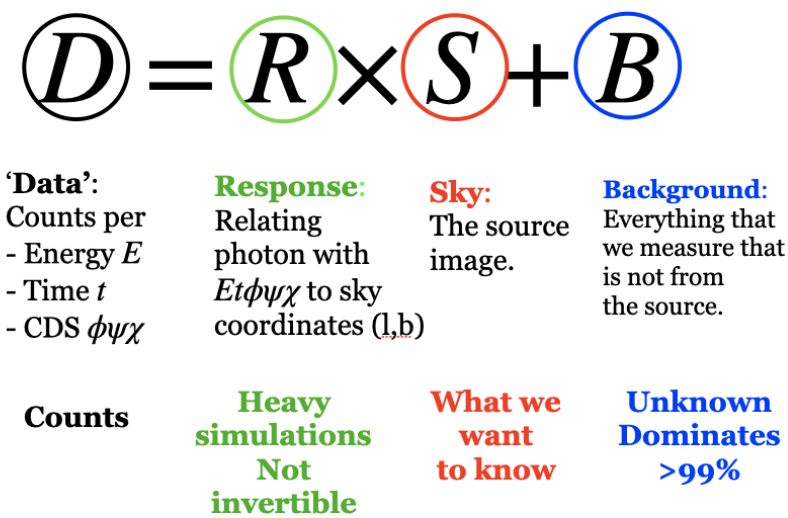


Fitting General Principle

Finally, we have all of our components, and now we can perform our analysis. When model fitting, we can free multiple spectral, polarization and location parameters. For simplicity, let's assume we will only fit for the amplitude of the source and background models that describe the data: D = **α**S + **β**B, shown schematically in the figure below, which is done for each energy bin and time bin independently. Through this procedure for spectral fitting and spatial model fitting, one can maximize the likelihood in the CDS.



If we don't know what the source should look like, instead of providing a sky model, we can be agnostic and perform image deconvolution. The data is a combination of the response convolved with the source sky distribution, and the addition of the background:



Since the response is non-invertible, we must use iterative deconvolutions to arrive at the sky distribution. In COSIpy-classic, we will introduce you to a modified Richardson-Lucy algorithm, which is a special case of the expectation maximization algorithm developed for COMPTEL's images of diffuse gamma-ray emission (Knödlseder et al. 1999). The Richardson-Lucy algorithm starts from an initial image, and iteratively compares the data in the CDS to the sky distribution convolved with the transpose of the response matrix R:

$$\hat{S}_{i+1} = \hat{S}_i \left(\frac{D}{\hat{S}_i \times R} \times R^{\dagger} \right)$$

This will evolve into the maximum likelihood solution. There are other image deconvolution techniques that will be used for COSI imaging, and those will be introduced in subsequent Data Challenges.

In order to get an intuition for how this equation is derived and to be understood, consider that we are trying to find the number of photons in an image pixel, given our model expectation $M=R \cdot S+B$. Here, M are the model counts, calculated from a linear combination of the instrumental background B and the sky morphology S, to which the image response function R is applied to convert from image space to data space.

In such a counting experiment, the measured data D given the model expectation M follows a Poisson distribution, so that the likelihood of measuring D photons is given by $P(D|M)=\prod(\exp(-M)M^D/D)!$. The product would be over all sky pixels in this case.

Taking the logarithm of the likelihood and plugging in the model expectation results in
$L \equiv \ln P(D|S,B) = \sum(-(R \cdot S + B) + D \ln(R \cdot B))$.

Assuming, for the moment, that the background is known, we can optimize the likelihood with respect to the wanted sky signal S, such that

$0 = \nabla_S L = -I \cdot R^T + (D \cdot R^T)/(R \cdot S + B) = (1 - D/(R \cdot S + B)) \cdot RT.$

Finally, this equation can be solved iteratively, (for each pixel simultaneously) by assuming a starting sky distribution (map) for S, similar to Newton's method for example, which will result in the RL equation above.

Next Steps

Now that you have a better feeling for the Compton telescope analysis process, at least in a general sense, you're ready to start on the analysis examples. First, we recommend you check out the detailed description of the simulations in the data_products directory. The easiest analysis is the spectral fitting, and we recommend you start there: spectral_fit. The Richardson-Lucy imaging is computationally intensive, and still largely hard-coded in this release, and thus we recommend you work through these notebooks second: imaging. If you have a workstation with a large memory allocation, we recommend you use that for these analyses - a personal laptop with only 16 GB of memory will be limiting. And then finally, we have also included some of the COSI flight data, specifically for analysis of the Crab as seen during the 2016 flight. After you've run through the spectral fitting and image deconvolution with the simulated data, you should be ready to analyze this real flight data: balloon data!

As mentioned previously, please don't hesitate to reach out to the COSI Data Challenge team if you have any questions, concerns, issues, or suggestions. Email Chris Karwin.

Data Challenge Simulations README

For the first Data Challenge, we wanted to give the users a basic look at COSI data analysis, so we chose 3 straightforward examples based on the COSI science goals:

- Extracting the spectra from the Crab, Cen A, Cyg X-1, and Vela
- Imaging bright point sources, such as the Crab and Cyg X-1
- Imaging diffuse emission from 511 keV and the 26Al 1.8 MeV gamma-ray line

For each of these examples, we have provided a detailed description of the simulated sources and data products here in the data_products directory. Each of the sources was simulated at 10x the astrophysical flux. Having a strong signal simplifies the analysis and allows us to focus on the workflow of the procedures. The COSI SMEX mission is expected to be 50x more sensitive than the balloon-borne mission

The simulations were all performed in MEGAlib, with an accurate mass model of the COSI Balloon instrument. The COSIBalloon.9Detector.geo.setup model, which accounts for the failure of three GeD detectors at different times during flight. Each of the continuum simulations was performed for 100 keV – 10 MeV, and an energy range selection of <5 MeV was used in MEGAlib's mimrec event selection tool.

Data Products:

We have included many combinations of the source simulations and background to allow for flexibility and further testing with these files. All files are either the .npz zipped numpy array format for the CDS-binned response matrix or background model, or the MEGAlib photon list .tra.gz format. MEGAlib was used to perform all of these simulations, and the source models are described in detail below.

Within this directory, there is 1 background model and 3 response matrices.

- Scaled_Ling_BG_1x.npz: background model generated from C. Karwin's scaled 1x Ling background simulation
- Continuum_Response.npz: 6º response used for spectral analysis and imaging continuum sources
- 511keV_imaging_response.npz: 6º imaging response required for RL imaging of Galactic positron annihilation
- 1809keV_imaging_response.npz: 6º imaging response required for RL imaging of Galactic Al-26

There is the full-sky simulation with background and all of the sources combined:

- DC1_combined_10x.tra.gz: 4 point sources with 10x flux (Crab, Cyg X1, Cen A, Vela), 511 kev & Al-26 lines, 1x Ling background

There is each of the sources individually with the background:

- Point_sources_10x_BG.tra.gz: 4 point sources with 10x flux (Crab, Cyg X1, Cen A, Vela) and 1x Ling background
- Crab_BG_10x.tra.gz: Crab with 10x flux and 1x Ling background
- CygX1_BG_10x.tra.gz: Cyg X1 with 10x flux and 1x Ling background
- CenA_BG_10x.tra.gz: Cen A with 10x flux and 1x Ling background
- Vela_BG_10x.tra.gz: Vela with 10x flux and 1x Ling background
- GC511_10xFlux_and_Ling.inc1.id1.extracted.tra.gz: 511 emission with 10x flux with Ling background
- Al26_10xFlux_and_Ling.inc1.id1.extracted.tra.gz: Al26 emission with 10x flux with Ling background

Each of the sources is also included without background:

- Point_sources_10x.tra.gz: 4 point sources with 10x flux (Crab, Cyg X1, Cen A, Vela)
- Crab_only_10x.tra.gz: Crab with 10x flux
- CygX1_only_10x.tra.gz: Cyg X1 with 10x flux
- CenA_only_10x.tra.gz: Cen A with 10x flux
- Vela_only_10x.tra.gz: Vela with 10x flux
- GC_511_10xFlux_only.inc1.id1.extracted.tra.gz: 511 emission with 10x flux
- Al26_10xFlux_Only.inc1.id1.extracted.tra.gz: Al26 emission with 10x flux

And finally, there is 1 background simulation (this is not required for any analysis, but is included here for posterity):

- Scaled_Ling_BG_1x.tra.gz: C. Karwin's scaled 1x Ling background simulation

All files have the same start and stop time, in unix time:

start time: 1463443400.0 s
stop time: 1467475400.0 s
total time: 4032000.0 s = 46.67 days

This corresponds to May 17 2016 00:03:20 GMT to Jul 02 2016 16:03:20 GMT, covering the full COSI Balloon flight in 2016.

As described in the main cosi-data-challenge-1 README, these files are stored on Git's Large File Server, and one needs to install git-lfs to access them.

Science Background

Point Sources

There are four bright point sources that are included in these simulations: the Crab nebula, Cygnus X-1, Centaurus A, and Vela.

The Crab nebula is considered both a pulsar wind nebula (PWN) and supernova remnant. The PWN surrounds the Crab Pulsar, a rapidly rotating and magnetized neutron star in the Milky Way constellation Taurus. The supernova remnant was produced by SN 1054. The Crab entered COSI-balloon's field of view for only ~12 of the 46 days of the 2016 flight (Sleator 2019); the balloon remained largely in Earth's Southern Hemisphere and the Crab is more easily viewed from the Northern Hemisphere. Nevertheless, as the brightest persistent γ-ray source in the sky, the Crab is detectable in the balloon data and is simulated for the data challenge with 10x its true flux of 0.049 ph $cm^{-2}s^{-1}$ (100 keV-50 MeV).

Cygnus X-1 is a bright hard X-ray source in the Cygnus constellation of the Milky Way. It is believed to be a black hole in an X-ray binary system. Cygnus X-1 emits in COSI's bandpass as well, and like the Crab is simulated here at 10x its true flux of 0.041 ph $cm^{-2}s^{-1}$ (100 keV-50 MeV). This data challenge thus helps establish expectations for COSI-balloon observations of Cygnus X-1 during the 2016 flight.

Centaurus A is a galaxy in the constellation of Centaurus. Also called NGC 5128, Centaurus A has a supermassive black hole which emits radio waves, X-rays, and gamma-rays. It is simulated with 10x its true flux of 0.0036 ph $cm^{-2}s^{-1}$ (100 keV-50 MeV) for the data challenge analyses.

The Vela pulsar is simulated based on a power law extrapolation of the corresponding Fermi-LAT source (4FGL J0835.3-4510) to lower energy. We have verified that the extrapolation is consistent with the observations from OSSE+COMPTEL+EGRET, e.g. see Pavlov+01. It is fainter than the other three sources but is included as a source which could be observed by an instrument like COSI-balloon with more observation time.

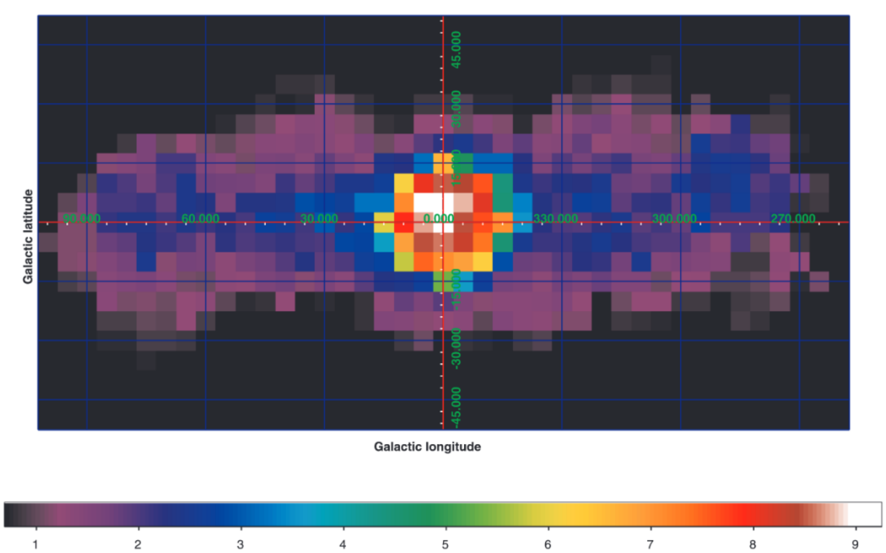
Positron Annihilation at 511 keV

A strong 511 keV signal emanating from the direction of the Galactic Center (ℓ=0◦,b=0◦) was first discovered in the 1970s on a series of balloon missions (Johnson III et al. 1972, Leventhal et al. 1978, Johnson III & Haymes 1973, Haymes et al. 1975, Ling et al. 1977, Albernhe et al. 1981). Despite decades of observation since the initial measurements, several questions remain regarding the nature of this abundant positron-electron annihilation. As MeV gamma-ray instruments, COSI-balloon and the COSI satellite are uniquely equipped to study this signal, which traces one of the biggest unsolved mysteries in gamma-ray astrophysics.

1. What is the underlying nature of the emission morphology?

The 511 keV image from the INTEGRAL SPI satellite (Bouchet et al. 2010, shown below as a significance map) reveals two seemingly distinct components: an extended "disk" which traces the Galactic Plane and a central, bright "bulge" around the Galactic Center. The notable difference between these structures is not understood. The extended disk emission may suggest that positrons are propagating away from and annihilating at a distance from their production sites, thereby smearing out the emission into a diffuse presentation. It is also possible, however, that there could be a collection of point-like sources emitting positrons which annihilate close to their progenitors; it is the collection of these sources together which may form a total diffuse structure. However, no point source of positrons has been detected. It is likely that positrons propagate away from their production sites and slow down to low enough energies to form positronium.

Imaging with fine angular resolution can help disentangle this dichotomy. The detection of an individual point source, for example, could revolutionize our understanding of the morphology. High-resolution spectroscopy is also critical to understanding the transport and eventual annihilation sites of these positrons: a significant ortho-Positronum (o-Ps; ≤511 keV) component of the spectrum would suggest annihilation in colder regions of the interstellar medium (ISM). A stronger signature at the line energy of 511 keV (para-Positronium, p-Ps) indicates annihilation in warmer regions of the ISM. Data from SPI currently favor the latter scenario, though additional measurements of this o-Ps to p-Ps fraction are necessary.



2. Where are all of these positrons coming from?

The bulge emission exhibits a positron annihilation rate of 10^{43}e+s^{-1}. Positrons from the β+ decay of nucleosynthesis products may account for the ~10^{42}e+s^{-1} in the disk and some of the bulge, but the origin of the remaining positrons in the bulge is unknown. Important to note is that there is no lack of potential positron sources; rather, there are too many possible sources to explain the emission! Positrons are readily created in a wide variety of astrophysical objects and processes. Stellar flares, massive stars, supernovae (core-collapse and Type Ia), classical novae, and neutron star mergers all synthesize radioactive isotopes which can β+ decay. Secondary interactions with cosmic rays produce positrons, positrons are created through pair creation channels in strong photon and magnetic fields, evaporating black holes may potentially produce positrons, dark matter density profiles may trace positron annihilation,...and more.

Ultimately, we can boil the "positron puzzle" down to uncertainty around the *source, transport, and sink* (the annihilation itself) of these particles.

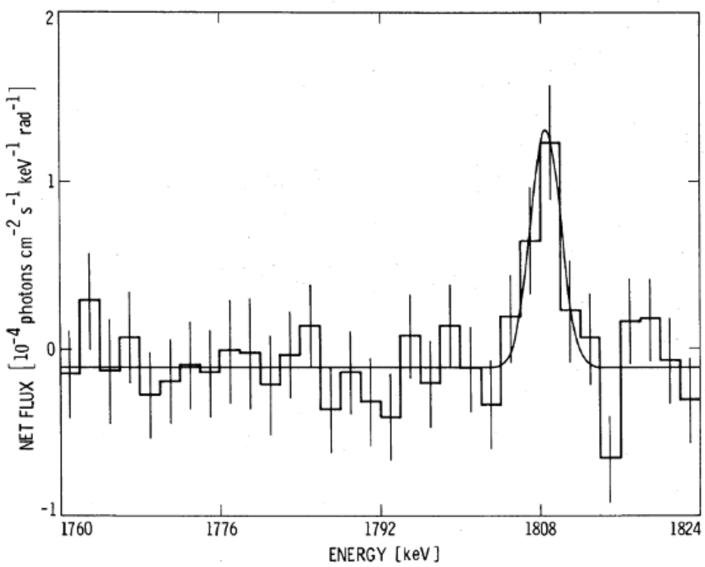
The COSI-balloon flight in 2016 detected the 511 keV signature of the positron puzzle with 7.2σ significance (Kierans et al. 2020). It also clearly imaged the bright bulge emission near the Galactic Center (Siegert et al. 2020). Both measurements are consistent with those from SPI and indicate additional extended emission.

In this data challenge, you will image the Galactic 511 keV emission, simulated at 10X its true flux (10X flux = 1.1×10^{-2} ph $cm^{-2}s^{-1}$) for robust statistics, as seen during the COSI-balloon flight in 2016. You should expect to see the bright bulge emission near the Galactic Center (but not the disk emission, which is too weak for COSI-balloon to see during its 46-day flight; SPI was able to detect the disk's ~1 ph/week with over a decade of observation time).

Al-26 Decay at 1.8 MeV

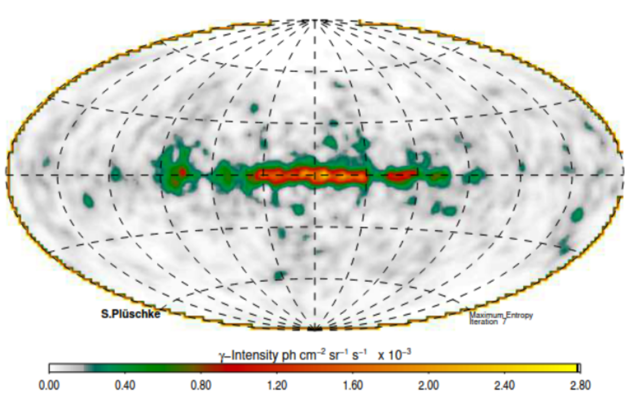
As MeV gamma-ray instruments, COSI-balloon and the COSI satellite are uniquely equipped to study the 1.809 MeV signature emission from this radioisotope, which traces stellar nucleosynthesis over millions of years.

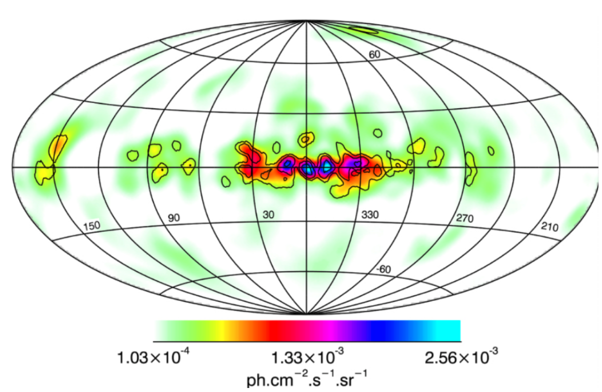
In the 1980s, NASA's High Energy Astrophysical Observatory (HEAO-3) satellite mission detected 1.809 MeV emission emanating from the direction of the Galactic Center Mahoney et al. 1984. This marked the discovery of Galactic Al-26. The spectrum is shown below.



Subsequent observations by the Compton telescope (COMPTEL) on-board NASA's Compton Gamma Ray Observatory (CGRO) yielded the first image of Al-26 emission (Oberlack et al. 1996, Oberlack 1997, Pluschke et al. 2001). Emission is concentrated in the Inner Galaxy (|ℓ| ≤ 30º, |b| ≤ 10º) with enhanced emission in regions of massive star activity, including Cygnus, Carina, and Vela.

The SPectrometer on INTEGRAL (SPI) largely corroborated the features seen in the COMPTEL image with over a decade of observation time from ESA's INTEGRAL satellite (Bouchet et al. 2015). Emission is concentrated in the Inner Galaxy with a reported flux of ~3.3×10^{-4} ph cm^{-2}s^{-1}. As in the COMPTEL image, there is enhanced emission in regions of massive star activity, including Perseus/Taurus, Cygnus/Cepheus, Carina, Vela, and Scorpius-Centaurus.





(**left**) COMPTEL 1.8 MeV image (Pluschke et al. 2001), (**right**) SPI 1.8 MeV image (Bouchet et al. 2015)

The COSI-balloon flight in 2016 measured the 1.809 MeV signature of Al-26 with 3.7σ significance, corresponding to ~106 Al-26 photons (Beechert et al. 2022). The reported Inner Galaxy flux of $(8.6\pm2.5)\times10^{-4}$ph cm^{-2}s^{-1} and line centroid of 1811.2±1.8 keV are consistent with results from SPI and COMPTEL within 2σ uncertainties. Future observations with the COSI satellite (significantly increased observation time, greater effective area at 1.8 MeV, better constraints on high-latitude emission, and finer angular resolution) will comprise an important comparison to the balloon measurement, which was the first measurement of Al-26 on a compact Compton telescope.

Furthermore, the COSI satellite's full-sky observations with fine angular resolution have potential to more closely study individual regions of massive star activity; in particular, resolving individual sites of emission within Cygnus is a promising goal of the mission. Detailed imaging and spectroscopic studies of the region may inform better understanding of the dynamics of Al-26 after it is produced and ejected from massive stars.

In this data challenge, you will image the Galactic Al-26 emission (traced by the DIRBE 240 μm image) as seen during the COSI-balloon flight in 2016. The flux is simulated at 10X the observed Al-26 flux (10X Inner Galaxy flux = 3.3×10^{-3} ph cm^{-2}s^{-1}; 10X total map flux = 1.2×10^{-2} ph cm^{-2}s^{-1}) for robust statistics. You should expect to see extended emission along the Galactic Plane, similar to that revealed by COMPTEL and SPI's 1.8 MeV images. The massive star regions of Cygnus, Carina, and Vela will not be as easily identifiable in this simulation of the balloon flight; for those, be sure to participate in the data challenge (and real data analysis) of the COSI satellite!

Source Models

Point Sources

There are four bright point sources that are included in these simulations: the Crab nebula, Cygnus X-1, Centaurus A, and Vela. The spectra and flux for each of these sources was determined from the literature. As explained above, we have used 10x the flux values for these sources in order to simplify the analysis. Please keep that in mind when you see the detections in this Data Challenge.

Crab:

(l,b) = (184.56, -5.78)

Spectral shape: Band function from [Jourdain et al. 2020]

Flux = 0.48977 ph/cm^2/s between 100 keV and 10 MeV

Cen A:

(l,b) = (309.52, 19.42)

Spectral shape: SED from [HESS+LAT collaboration 2018]

Flux: 0.03609 ph/cm^2/s between 100 keV and 10 MeV

Cyg X1:

(l,b) = (71.33, 3.07)

Spectral shape: SED from [Kantzas+21]

Flux = 0.40644 ph/cm^2/s between 100 keV - 10 MeV

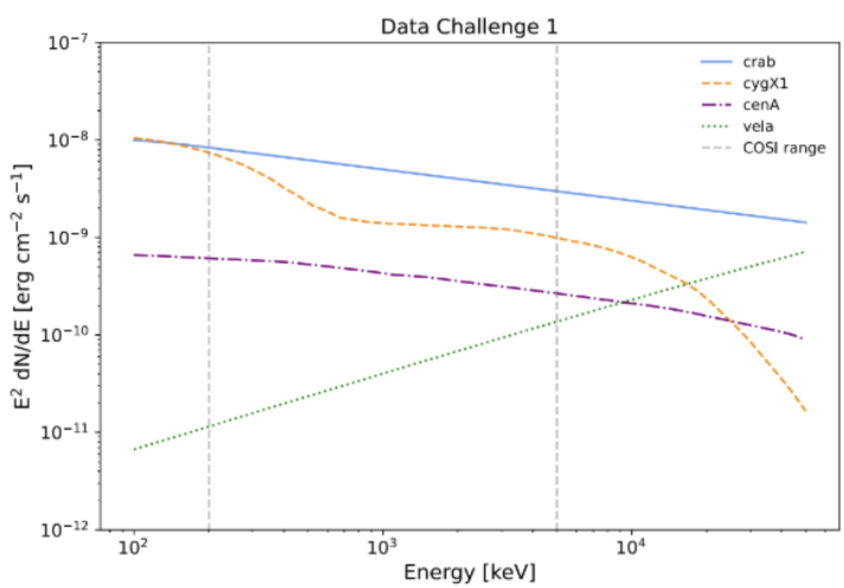
Vela:

(l,b) = (263.55, -2.79)

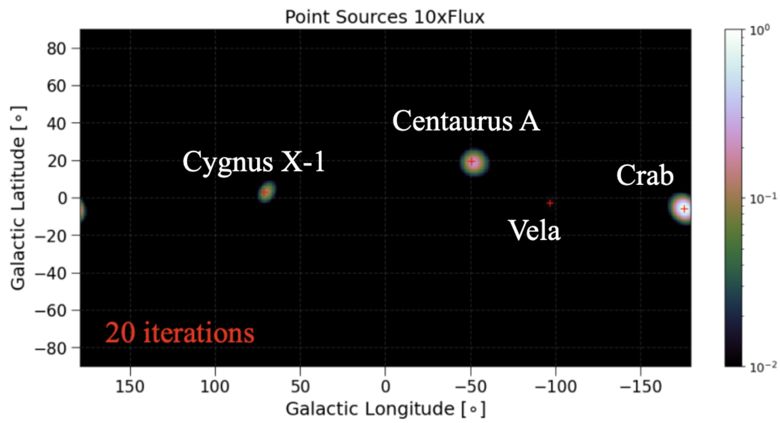
Spectral shape: Power law extrapolation of the Fermi-LAT Vela pulsar (4FGL J0835.3-4510), see Abdollahi+20

Flux = 0.00120 ph/cm2/s between 100 keV - 10 MeV

The input spectra for these point sources is shown below. Note that the plot shows the true spectra, whereas the flux values reported above are for the 10x simulations.



The simulations are run in MEGAlib's cosima tool, and then a list-mode image is created in mimrec to confirm the correct point source locations:

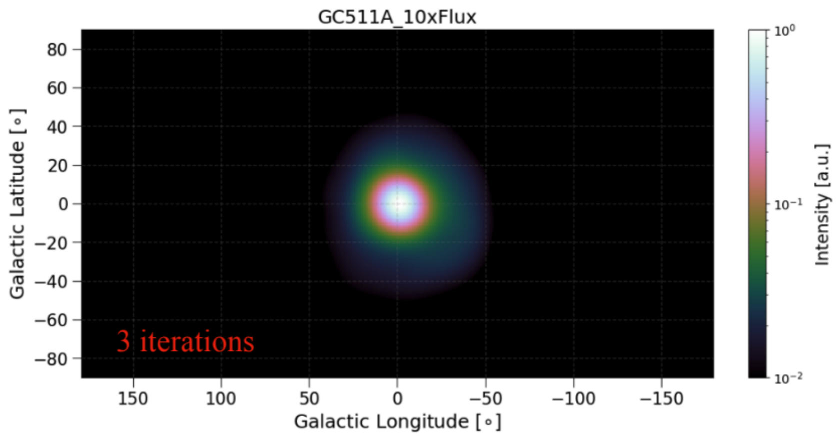


Positron Annihilation at 511 keV

The morphology of the 511 keV emission from positron annihilation is not well constrained. For this first Data Challenge, we have used the model defined in Knödlseder et al. 2005, where the emission was fit with a 2-D asymmetric Gaussian spatial model with the following parameters (the reported flux is the true value and not the 10x value):

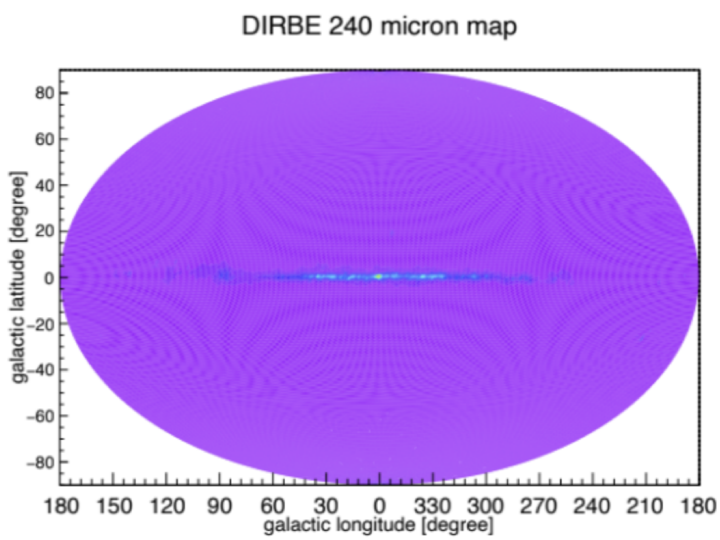
Quantity	Measured value
RMLR (DOF)	462.2 (5)
l_0	$-0.6° \pm 0.3°$
b_0	$+0.1° \pm 0.3°$
Δl (*FWHM*)	$8.1° \pm 0.9°$
Δb (*FWHM*)	$7.2° \pm 0.9°$
511 keV flux (10^{-3} ph cm^{-2} s^{-1})	1.09 ± 0.04

The emission was simulated as a 511 keV mono-energetic source, and the image from mimrec confirms the extended emission in the Galactic Center.

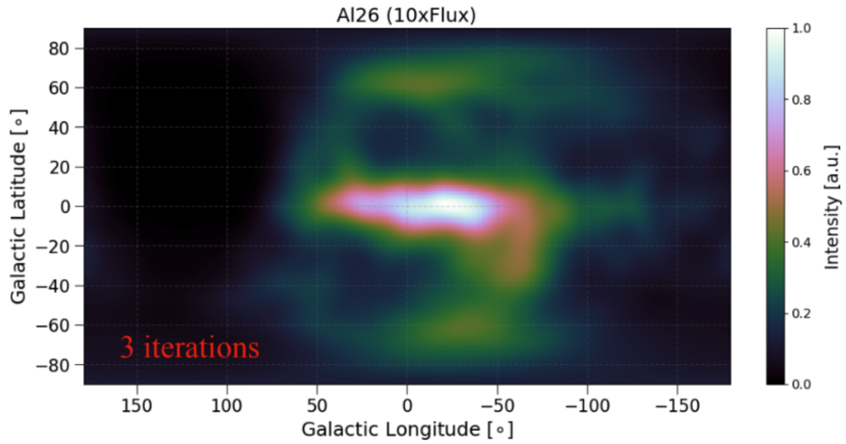


Aluminum-26 Decay at 1.8 MeV

The Diffuse Infrared Background Experiment (DIRBE) 240 um map has been shown to be a good tracer for the Al-26 emission, as measured by COMPTEL and INTEGRAL/SPI. We use that distribution as the spatial model for the Al-26 emission, and the inner Galaxy flux was normalized to 3.3×10−4 ph/cm2/s (Diehl et al. 2006). This model is described further in Beechert et al. 2022.



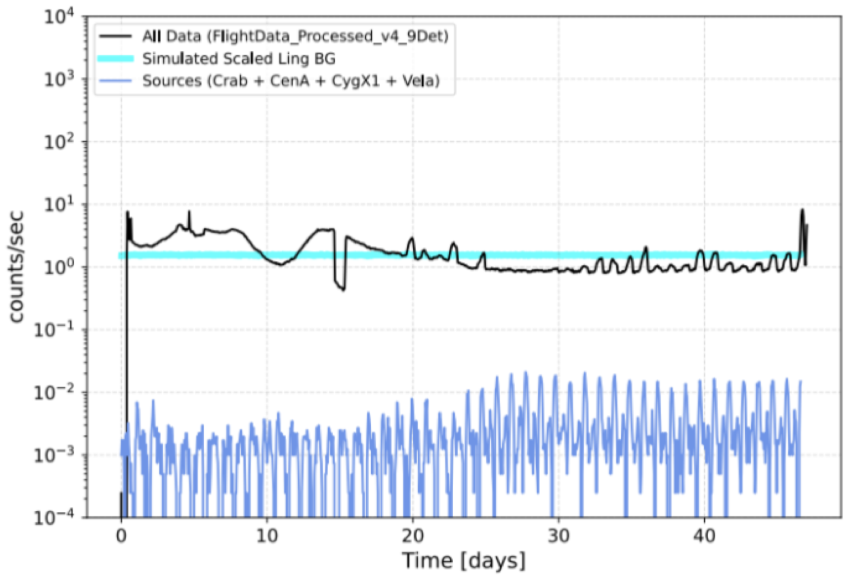
The emission was simulated with a 1.8 MeV mono-energetic source, and the image from mimrec confirms the extended emission along the Galactic disk.



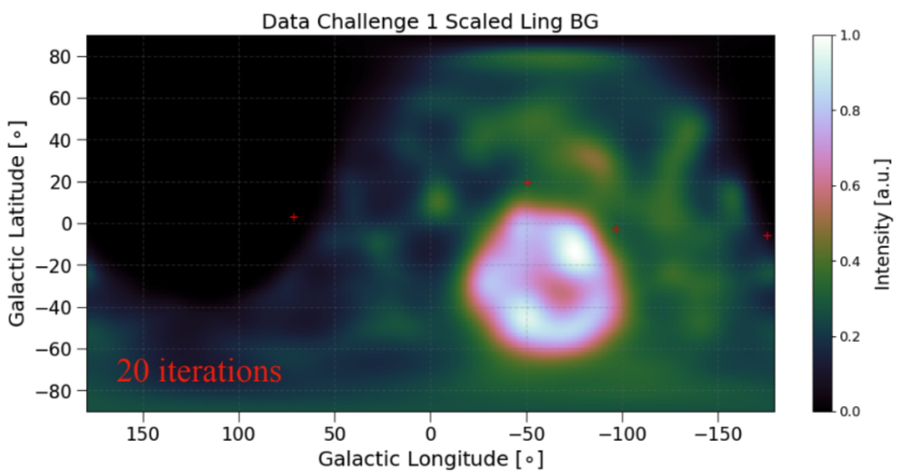
Note that the list-mode imaging method employed in MEGAlib is not optimized for diffuse sources, and the extra structure out of the Galactic plane is an imaging artifact.

The background radiation model used for the Data Challenge is based on the semi-empirical model from Ling 1975, which has been used for all COSI balloon analyses. The model describes the angular and atmospheric depth dependance of gamma-rays within 0.3 – 10 MeV and includes 3 primary gamma-ray sources: continuum emission (Bremsstrahlung from cosmic ray interactions in the atmosphere), 511 keV line component (from electron-positron annihilation in the atmosphere, and cosmic diffuse component (down-scattered gamma-rays from the extragalactic background and Galactic plane). The Ling model requires a description of the atmosphere's density, interaction depth, and mass absorption coefficients, which are obtained through the NRLMSISE-00 model. An altitude of 33.5 km was assumed for these models - altitude drops and changes in longitude and latitude were not taken into account for this simulation.

The amplitude of the Ling background was scaled so the total integrated background spectrum from simulation matched closely to what was measured during flight, as can been seen in the figure below. The flight data ("All Data" label) shows a time-variable background count rate that was influenced by the geomagnetic cutoff, and balloon altitude drops. The cyan line shows the scaled Ling background model. As a reference, the count rate from the 1x flux point sources are shown, and the signal is at most a few percent of the background count rate.

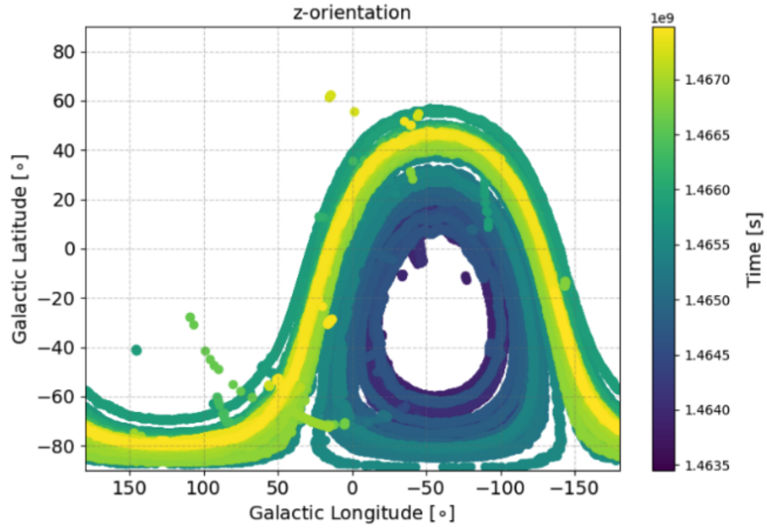


An image of the background simulation traces the exposure map, since the orientation of the COSI Balloon in Galactic coordinates was included in the simulation:



Flight path and transmission

The source simulations include the real flight aspect information so that the balloon path and source exposure time is accurate. This can be seen in the below plot showing the Galactic longitude and latitude of the zenith direction of the COSI balloon as a function of time.



Furthermore, the transmission probability of the source photons in the atmosphere are calculated for each instance of the simulation. The probability of transmission is taken at a constant altitude of 33 km, and is shown as a function of zenith angle and energy in the below figure.